

BUSINESS INTELLIGENCE · IIB423T-1 · CERTAMEN 1 (C1)

Respiratory Emergency Visits in Chile (2020-2024)

Full Exploratory Data Analysis, Five Cs Quality Audit, and Capacity Planning for the Winter Campaign

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Operational Context & The Winter Campaign Decision

- **The Public Health Concern:** Recurrent seasonal surges in acute respiratory infections (RSV, Influenza A/B, Adenovirus, COVID-19) place severe pressure on public hospital emergency departments and primary care.
- **Target Users:** Directorate for Health Care Network Management (DIGERA), Health Service directors, and emergency leaders across SAPU, SAR, SUR, and Hospitals.
- **The Decision to Inform:** Timing the activation of Winter Campaign measures: scheduling clinical shifts (physicians, nurses, kinesitherapy), phased critical bed conversions, and primary care triage diversion.
- **Core KPI & Formula:** $\text{Respiratory Share (\%)} = 100 \times \frac{\text{Total Respiratory Visits (IdCausa = 2)}}{\text{Total Emergency Visits (IdCausa = 1)}}$
- **Critical Methodological Boundary:** This measures recorded public-network consultation activity. It does NOT estimate individual clinical risk, disease severity, bed occupancy, or true population incidence.

Selection from W1 and the C1 Reproducibility Correction

- **Why Candidate A (Respiratory) was Selected:** Continuous, high-frequency weekly series (262 dated weeks across 2020–2024) with complete reporting, directly supporting seasonal monitoring.
- **Why Candidate B (REM-P2 Child Nutrition) was Deselected:** Audit uncovered severe reporting collapse in June 2020 (only 83 facilities reported due to lockdown vs >1,950 normally); only 10 reporting points in 5 years; >59% age/sex breakdown missing; code changes in 2023.
- **The W1-to-C1 Reproducibility Correction:** While W1 reported 5-year aggregates, its packaged reduced extract contained only sample peak weeks in 2023–2024. For C1, an internal teammate review expanded the extract to EVERY date across all 5 years (2020–2024).
- **Auditable Extract Scope:** 2,169,824 uncleaned source rows (~85x larger than W1's extract) while maintaining an efficient 19 MB compressed archive size for seamless verification.

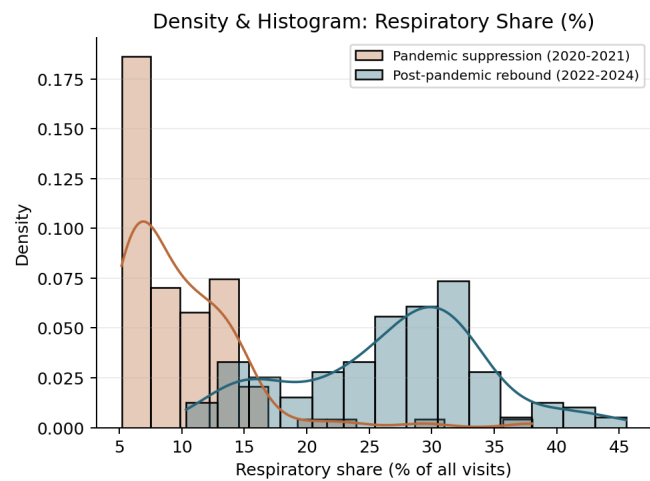
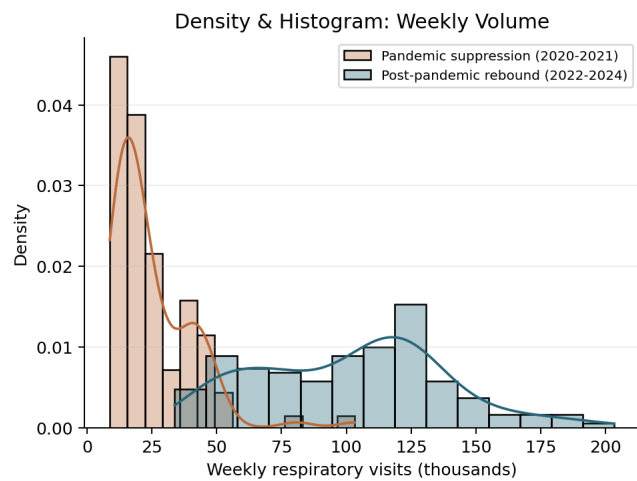
Application of the Seven Steps (Class 03 Framework)

- **1. Decision:** Proposed winter planners review weekly demand trajectory to determine when to activate phased staffing and critical bed contingency plans.
- **2. Success Criteria:** Construct a reproducible 2020–2024 weekly baseline with audited coverage, stable definitions, and verified care-setting coupling.
- **3. Business Question:** How did the weekly volume and share of respiratory emergency visits vary in 2020–2024 across care settings and age bands?
- **4. Care Process:** Daily emergency consultation records (DAU/SADU) logged at health facilities and transmitted to DEIS.
- **5. Entities & Events:** Entity: Health facility. Event: Recorded emergency visit. Observation unit: Aggregate facility × date × cause count (NOT unique patients).
- **6. Representation:** Candidate key: (source_year, IdEstablecimiento, fecha, IdCausa). Clean case derives Sunday week starts from parsed dates.
- **7. Measurement:** Respiratory Visits = Total(IdCausa=2); Respiratory Share (%) = Total(IdCausa=2) / Total(IdCausa=1) × 100 within identical periods/units.

Data Profiling & The Five Cs Quality Audit

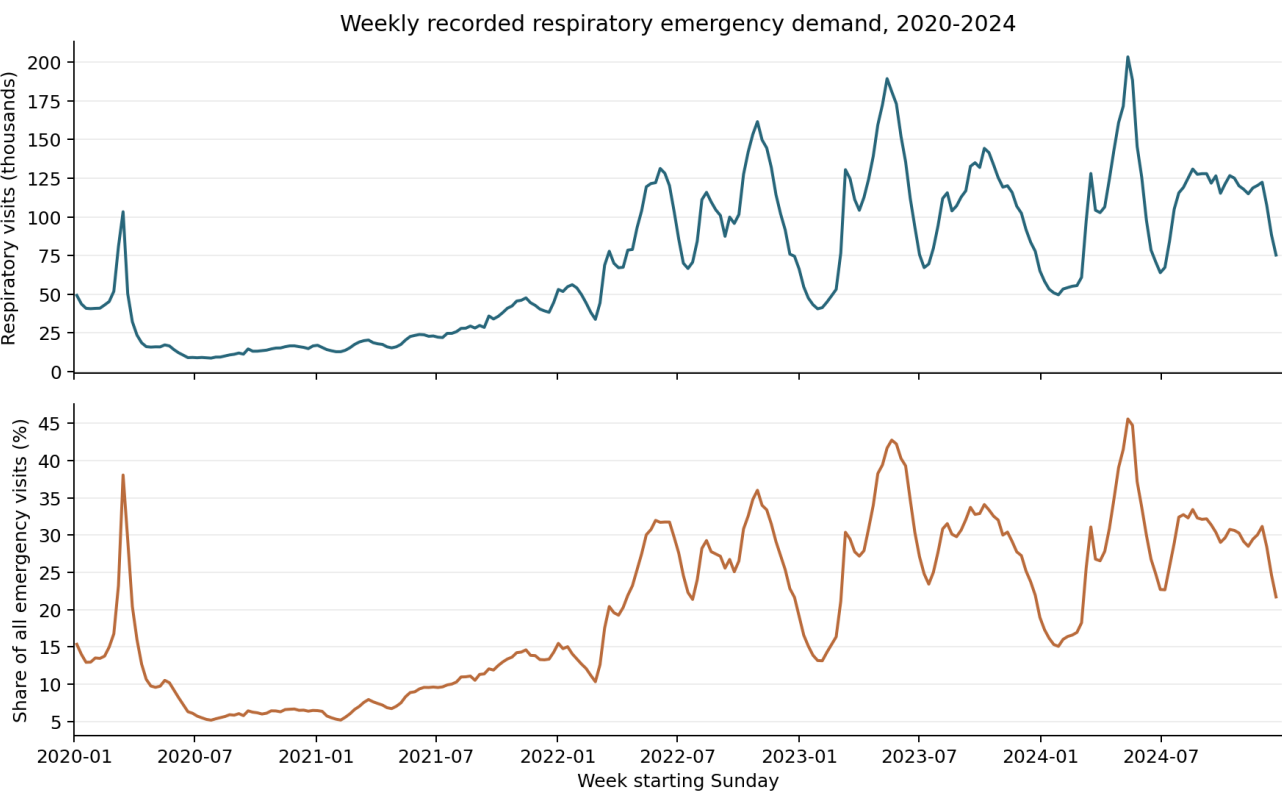
- **Clean:** Checked 2,169,824 rows: zero missing key/measure cells; zero negative values; age sums match Total 100%. Exactly 2 identical duplicate rows in 2023 removed only from derived tables (-13 all visits, -0 respiratory). Valid zero-count rows retained.
- **Consistent:** IdCausa 1 and 2 maintain stable definitions across 2020–2024. Parsing dates into Sunday week starts resolves calendar boundary collisions where the same (year, semana) label mapped to two disjoint weeks.
- **Conformed:** Harmonized schemas across annual DEIS archives. No external geographic join is claimed: 2020–2022 lack direct comuna/region fields, and unvalidated joins risk creating duplicate or unmatched rows.
- **Current:** The 2020–2024 series is appropriate for historical baseline analysis. It cannot directly guide live 2026 shift decisions without an automated operational feed.
- **Comprehensive:** Covers the entire reporting public emergency network (>600 facilities annually). Does not cover private clinic care or self-treated community cases.

Histograms & Kernel Density Estimation (KDE)



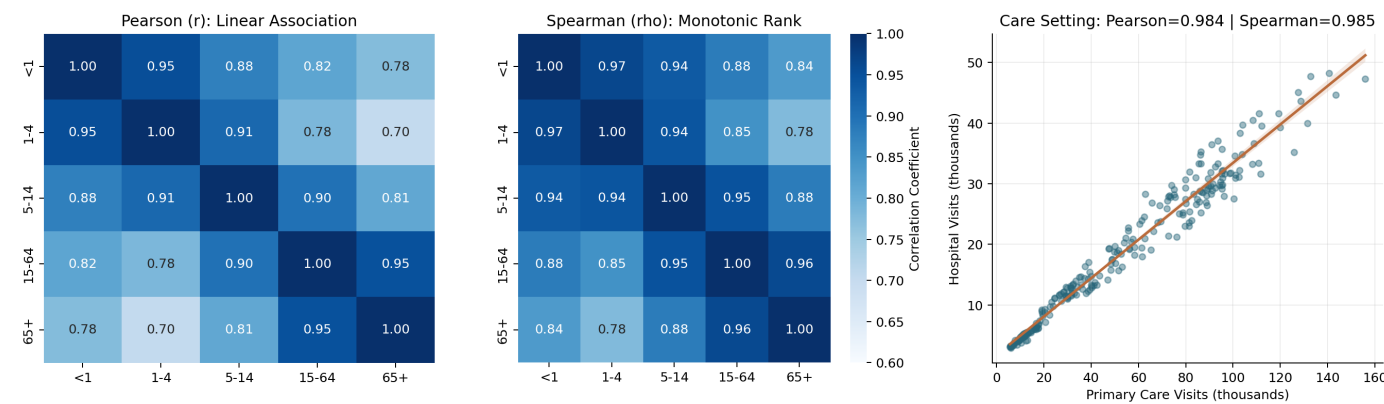
- **Parametric Non-Normality:** Weekly volume and share distributions exhibit high positive skewness and heavy right tails during epidemic surges; normality assumptions fail.
- **Pandemic Suppression (2020-2021):** Unimodal, tightly bounded distribution centered at ~10% share and <50k weekly visits due to school closures and mask mandates.
- **Post-Pandemic Rebound (2022-2024):** Multimodal distribution with broad dispersion reaching 45.56% share and 203,353 visits/week during winter peaks.
- **Operational Implication:** Planning models must account for regime shifts; assuming a single Gaussian distribution severely underestimates peak surge risk.

Weekly Demand Dynamics & Autumn Volume Peaks



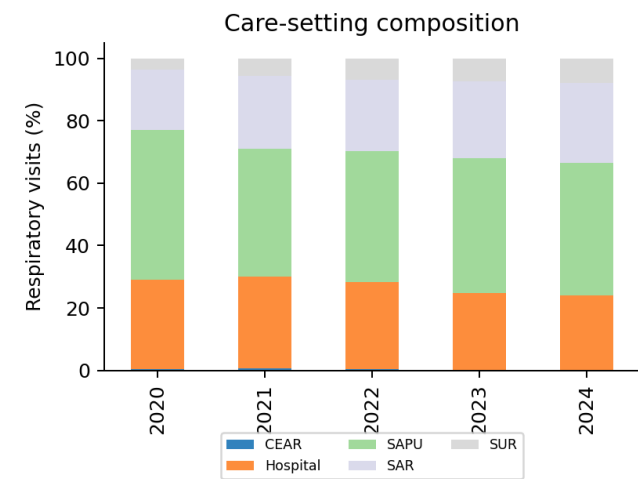
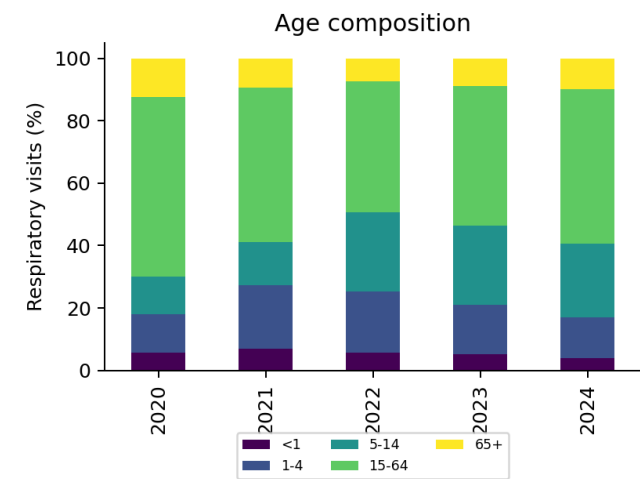
- **260 Complete Weeks Analyzed:** Edge weeks (4 and 3 days) excluded for strict cross-year peak comparability.
- **2023 Peak:** Week of 14–20 May: 189,297 respiratory visits (41.66% share).
- **2024 Peak:** Week of 12–18 May: 203,353 respiratory visits (45.56% share).
- **Timing Corroboration:** MINSAL's 24 May 2024 official campaign report recorded 45.4% week-20 share, broadly corroborating this archive's 45.56% timing.
- **Key Takeaway for Planners:** Peaks in recent years occurred in mid-May (late autumn), NOT in July/August (mid-winter). Preparedness reviews must begin early in April.

Pearson (Linear) vs. Spearman (Rank) Correlation Analysis



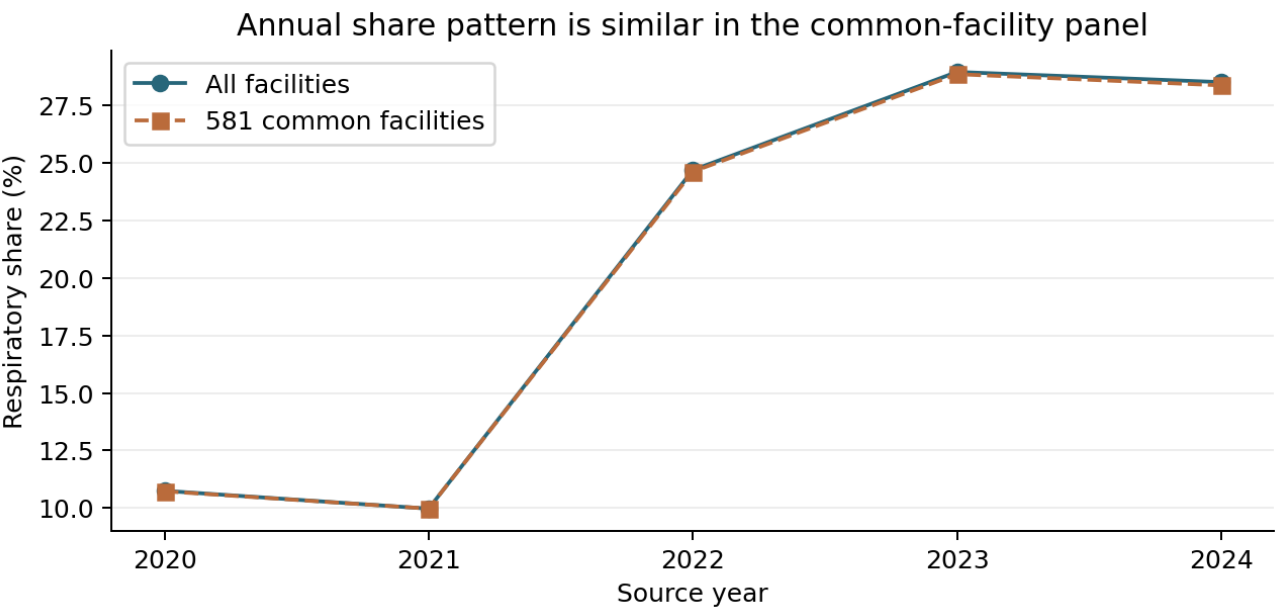
- **Linear vs Monotonic Association:** Pearson evaluates linear bivariate normality; Spearman evaluates monotonic rank ordering, robust against non-linear surges and outliers.
- **Age Cohort Synchronization:** Spearman is systematically higher than Pearson across all age groups (+0.012 to +0.074). Between <1 and 65+, correlation rises from $r = 0.775$ (Pearson) to $\rho = 0.835$ (Spearman).
- **Statistical Reason:** Different virus onset slopes (steep early RSV surge in infants vs delayed influenza in elderly) depress linear Pearson, but rank severity across weeks is strongly preserved.
- **Network Coupling:** Primary Care (SAPU/SAR) vs Hospital demand has Pearson $r = 0.9835$ and Spearman $\rho = 0.9848$. Triage and hospital pressure move in lockstep.

Care-Setting Distribution & Vulnerable Age Cohorts



- **Primary Care Shock Absorber:** SAPU, SAR, and SUR facilities handle between 71.08% (2020) and 75.87% (2024) of all recorded respiratory emergencies.
- **Pediatric Surge Dynamics:** Under-15 patients represented 30.05% of respiratory consultations in 2020, spiking to 50.71% in 2022 with school reopening, and settling at 40.72% in 2024.
- **Older Adult Demand:** Adults aged 65+ account for ~15-18% of visits, but represent the highest risk of hospitalization and ICU bed admission.
- **Design Consequence:** Monitoring cannot look only at hospitals. Primary care triage volume is the earliest and largest indicator of winter network strain.

Common-Facility Panel Check (581 Continuous Centers)



- **Compositional Bias Risk:** Total reporting facilities varied from 607 (2020) to 642 (2023). Did facility entry/exit create the observed respiratory surge?
- **The Stable Panel:** 581 facility IDs reported across ALL five years, covering 98.05% of all emergency visits in 2020 and 95.89% in 2024.
- **Empirical Result:** Respiratory share in the common-facility panel: 10.73%, 9.97%, 24.62%, 28.86%, 28.37% (virtually identical to full sample).
- **Conclusion:** The dramatic post-pandemic rebound is a genuine epidemiological and care-seeking reality, not an artifact of reporting panel turnover.

Project Continuity Beyond C1: Machine Learning & Dashboard

- **Future Supervised ML Task:** Forecast weekly respiratory emergency visits 2 completed weeks ahead (t+2) by care setting (SAPU/SAR vs Hospitals).
- **Information Boundary & Data Leakage:** Predictors restricted strictly to information available prior to the forecast week: lagged volume counts, calendar week indicators, and historical seasonal moving averages. No target week counts or post-outcome fields used.
- **Honest ML Feasibility Challenges:** With only 5 annual cycles and a strong regime shift in 2020–2021, standard time-series splits face high variance. Operational deployment requires an automated live feed. (No model trained in C1).
- **Future BI Dashboard Prototype:** Targeted at DIGERA and Health Service emergency coordinators.
- **Core Dashboard Views:** 1) Real-time weekly demand curve against 5-year historical confidence bands; 2) Pediatric (<1, 1-4) vs Geriatric (65+) composition; 3) SAPU/SAR absorption ratio; 4) Reporting facility freshness and completeness flags.

Individual Defense Roles & Methodological Anchors (Q&A;)

- **What This Analysis Demonstrates (F):** Defensible, auditable 5-year retrospective characterization of respiratory emergency demand, peak timing (May), age co-movement, and care-setting coupling in Chile.
- **What This Analysis Cannot Claim (H):** Does not estimate individual patient risk, true disease incidence, bed occupancy rates, triage wait times, or causal impacts of health campaigns.
- **Benjamín Pinto — Methodological Lead (G):** Processing 2.17M rows; 5 Cs quality audit; duplicate key resolution; date-derived Sunday week starts; statistical correlation analysis (Pearson vs Spearman); parametric vs nonparametric density.
- **Sebastián Herrera — Selection & Sensitivity Lead (G):** Candidate B deselection audit (REM-P2 coverage collapse in June 2020); 581 common-facility panel sensitivity; care-setting breakdown (SAPU vs Hospital); dashboard design & operational framing.